LD + EmbeddingGemma-WalkLM + SAGN on `ogbn-products`: A Ten-Seed Exploratory Study

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Abstract

We study a language-augmented node-classification pipeline for the Open Graph Benchmark `ogbn-products` dataset. The method concatenates released Label Deconvolution (LD) BERT features, graph-walk text embeddings from `google/embeddinggemma-300m`, and the original OGB features, then trains a four-stage SAGN/SLE predictor and applies Correct & Smooth (C&S). Across seeds 0–9, the recorded final validation accuracy is 0.940010 ± 0.000299 and test accuracy is 0.874753 ± 0.000487, using unbiased sample standard deviation. The artifact audit recovered the exact embedding command, input checksums, training outputs, and parameter accounting. It also found that test scores were viewed during encoder and C&S selection. We therefore report this as an exploratory result and do not claim that it is eligible for the OGB leaderboard without an explicit ruling or a clean validation-only rerun.

1 Dataset and evaluation

We use the official `ogbn-products` split and the OGB evaluator (OGB 1.3.6). After the preprocessing used by SAGN, the graph contains 2,449,029 nodes and 123,718,280 directed edges, with 196,615 training, 39,323 validation, and 2,213,091 test nodes over 47 classes. Ten independent runs use seeds 0 through 9. We report the arithmetic mean and unbiased sample standard deviation ($n-1$ denominator) of the raw per-seed evaluator values.

2 Method

2.1 Input representation

For every node, we concatenate the following tensors in a fixed column order:

Component	Dimension	Origin
LD BERT hidden state	768	Released `products_sagn/hidden_state.pt` from the LD authors
EmbeddingGemma-WalkLM	768	Masked-mean pooled, normalized embeddings of attributed random walks
Raw OGB feature	100	Original `ogbn-products` node feature
Concatenated input	1,636	Order: LD, EmbeddingGemma-WalkLM, raw

The LD feature is external released data. The product text and pretrained EmbeddingGemma model are also external data, so the method must be submitted with *External data: Yes*.

2.2 EmbeddingGemma-WalkLM extraction

Directed random walks are sampled independently for every node with seed 42. Four walks of length four are generated. Each visited product contributes the first 160 characters of its released text, and visited nodes are separated by the token "cites." Text is tokenized to at most 160 tokens. The final hidden state of `google/embeddinggemma-300m` is mean pooled with the attention mask and L2 normalized. The encoder is not fine-tuned. Batch size is four. The recovered command is:

```
PYTHONPATH=. python -u walklm.py \
  --dataset ogbn-products \
  --hf_model google/embeddinggemma-300m \
  --encoder_type auto_model --pooling mean \
  --normalize_embeddings --mlm_epochs 0 --epochs 1 \
  --batch_size 4 --num_walks 4 --walk_length 4 \
  --max_length 160 --max_text_chars 160 \
  --use_raw_features --save_embeddings
```

The artifact is a PyTorch dictionary containing `walklm_x` and `raw_x`. Its recorded size is 8,503,030,973 bytes and SHA-256 is `fef45b647df5155bf17505508cb08ee5a164ad9b33b69910ad1b437109721c90`. A standalone, label-free preservation of this path is provided in `scripts/generate_embeddinggemma_features.py`.

2.3 SAGN/SLE and post-processing

The final predictor uses two feature-propagation hops, nine label-propagation hops, 512 hidden channels, one attention head, learning rate 0.001, batch size 50,000, and dropout values 0.5/0.4/0.2/0.5 for model, attention, input, and label paths. Four self-training stages run for 300, 100, 100, and 100 epochs with thresholds 0.95, 0.95, and 0.999. Feature propagation is split into four CPU-memory chunks. Each seed is post-processed with 50 correction and 50 smoothing iterations, DAD normalization, correction alpha 0.30, smoothing alpha 0.65, and scale 25.

Only training labels are passed to SAGN/SLE label features and C&S. Validation labels are used for early stopping/evaluation. The implementation does not feed validation or test labels into propagation.

3 Results

Evaluation	Train	Validation	Test
Stage 3 before C&S	0.958059 ± 0.000949	0.938901 ± 0.000271	0.872624 ± 0.000492
After C&S	0.984537 ± 0.000081	0.940010 ± 0.000299	0.874753 ± 0.000487

The final test mean is 0.001053 above the 0.8737 public leaderboard entry visible on September 3, 2026. This numerical comparison is not an official rank and does not override the protocol limitation in Section 6.

4 Parameter accounting

Following OGB's multi-stage counting rule, we conservatively sum all learned feature-generation and prediction components:

Component	Parameters
LD `bert-base-uncased` encoder	109,482,240
LD SAGN/SLE head and aggregation weight	1,154,656
EmbeddingGemma encoder	307,581,696
Final 1,636-input SAGN/SLE predictor	5,010,014
Correct & Smooth	0
Total	423,228,606

The first two rows sum to OGB's published 110,636,896 parameter count for the LD+SAGN products entry. The final SAGN count is the runtime count printed by every preserved training log.

5 Reproducibility

The repository pins SAGN, LD, and WalkLM revisions; includes a validated patch for the 1,636-dimensional feature integration; archives exact training and C&S commands; preserves per-seed evaluator outputs; and links a release with all ten final prediction tensors, training logs, and checkpoint hashes. Both external tensor byte sizes and SHA-256 checksums are recorded in the input manifest. The LD tensor is 7,523,417,875 bytes with SHA-256 `fb1309cb73d4b1199b3eef342d6d3dc7ce2d2e38c57d50543fd8da0c8bbd6ec7`.

6 Protocol audit

The experiment history shows two forms of test-aware selection. First, several text encoders were screened by reporting OGB test-split accuracy on a 725,745-node subset before EmbeddingGemma was selected for the full graph. Second, C&S candidates were explicitly compared and sorted by test accuracy. The recorded setting (0.30, 0.65, 25) was retained during this exploration. This conflicts with the OGB rule that test labels are for final evaluation.

The repository therefore marks `leaderboard_eligible_as_recorded` as false and includes a validation-only C&S grid and locking utility for future work. That utility does not retroactively erase prior test feedback. A submission should proceed only after disclosing this history to OGB and receiving a favorable ruling, or after developing a clean, pre-registered method whose choices use validation results only.

7 Conclusion

The preserved ten-seed experiment is numerically competitive and substantially more reproducible than the original ad hoc run. Its scientific value is as an exploratory result and artifact package. The remaining barrier to an official leaderboard claim is experimental protocol, not the arithmetic or checkpoint record.

References

1. W. Hu et al. Open Graph Benchmark: Datasets for Machine Learning on Graphs. NeurIPS 2020. https://ogb.stanford.edu/
2. C. Sun et al. Scalable and Adaptive Graph Neural Networks with Self-Label-Enhanced Training. https://github.com/skepsun/SAGN_with_SLE
3. Z. Shi et al. Label Deconvolution for Node Representation Learning on Large-scale Attributed Graphs against Learning Bias. https://arxiv.org/abs/2309.14907
4. Google. EmbeddingGemma documentation. https://ai.google.dev/gemma/docs/embeddinggemma
5. Experiment code and artifacts. https://github.com/MeghanaShanthappa/ogbn-products-leaderboard_walklm